

Пробная олимпиада ISTCiM – 1

Time allowed: 4 hours

14 сентября 2026 года

A. Algebra and combinatorics. Find a non-zero polynomial $P(x, y)$ such that $P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$ for every real number a . (Here $\lfloor t \rfloor$ is the greatest integer not exceeding t .)

C. Calculus and mathematical analysis. Let $x_0 = 1$ and $x_{n+1} = \ln(e^{x_n} - x_n)$ for $n \geq 0$. Prove that the series $x_0 + x_1 + x_2 + \dots$ converges, and find its sum.

E. Equations and inequalities. Functions f, g, h are differentiable on an open interval around 0 and satisfy

$$f' = 2f^2gh + \frac{1}{gh}, \quad g' = fg^2h + \frac{4}{fh}, \quad h' = 3fgh^2 + \frac{1}{fg},$$

with $f(0) = g(0) = h(0) = 1$. Find an explicit formula for $f(x)$, valid in some open interval around 0.

G. Geometry and linear algebra. Let $A \in M_{m,n}(\mathbb{C})$ and $B \in M_{n,m}(\mathbb{C})$. Prove that

$$\text{rank}(I_m - AB) - \text{rank}(I_n - BA) = m - n.$$

P. Set theory and probability. Let n be a positive integer. Put $a_{n,0} = 1$, and for $k \geq 0$ choose $m_{n,k}$ uniformly at random from $\{1, \dots, n\}$ and set

$$a_{n,k+1} = \begin{cases} a_{n,k} + 1, & m_{n,k} > a_{n,k}, \\ a_{n,k}, & m_{n,k} = a_{n,k}, \\ a_{n,k} - 1, & m_{n,k} < a_{n,k}. \end{cases}$$

Let $E(n)$ be the expected value of $a_{n,n}$. Determine $\lim_{n \rightarrow \infty} E(n)/n$.